Deep Learning for Minor Motion Magnification: Principles, Applications and Challenges

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fake detection. Across these domains, we identify persistent challenges: limited real-world annotated datasets, sensitivity to illumination and sensor noise, edge distortion, insufficient physical interpretability, weak reproducibility and limited evidence for real-time deployment. Finally, we outline future directions, including physically constrained learning, self-supervised and unsupervised training, multimodal sensing, benchmark construction, uncertainty-aware evaluation and lightweight models for practical systems. By connecting classical principles with current neural approaches, this review aims to provide a coherent taxonomy and a technically grounded roadmap for researchers developing robust and reusable minor motion magnification methods.

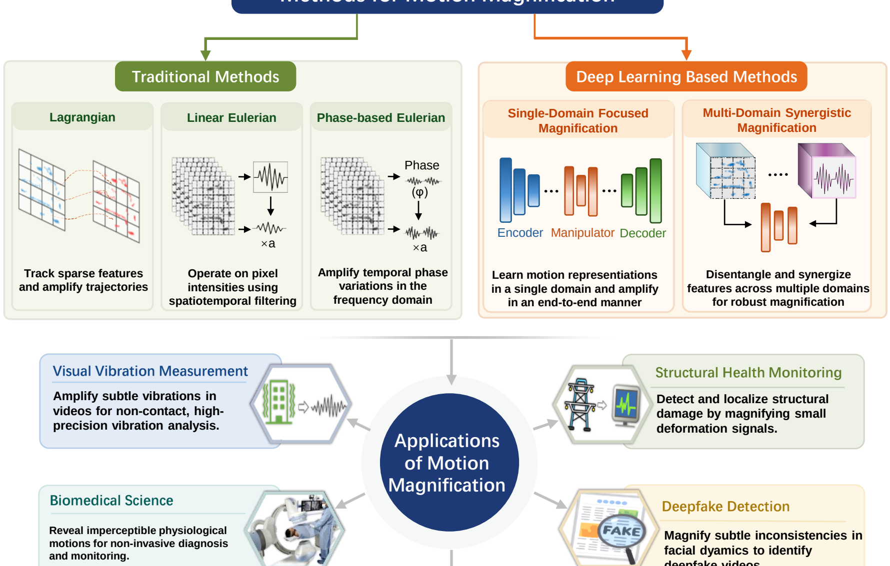
Keywords: Minor motion magnification, Deep learning, Video enhancement, Visual computing, Structural health monitoring

1 Introduction

Minor motion is a fundamental phenomenon observed across multiple scales, from effective under controlled conditions, they exhibit notable limitations in complex scenarios, such as sensitivity to noise, limited capability in handling non-rigid motion, and restricted amplification factors [23]. In addition, these methods typically require manual parameter tuning, which constrains their adaptability across diverse application environments.

With the rapid advancement of deep learning, data-driven motion magnification methods have attracted increasing attention. By learning motion representations directly from data, these approaches enable end-to-end optimization and improved robustness. Convolutional and transformer-based architectures, for example, have demonstrated strong performance in modeling non-linear motion, suppressing noise, and preserving structural details, thereby extending the applicability of MM to more complex real-world scenarios.

Despite these advances, existing reviews remain limited in several aspects. First, most surveys predominantly focus on classical signal-processing pipelines, with comparatively limited attention given to the systematic organization of deep learning-based methods. Second, deep learning approaches are often treated as monolithic end-to-end models, without explicitly analyzing how different architectural components (e.g., convolutional operations, attention mechanisms, or multi-domain


Methods for Motion Magnification
Traditional Methods
Lagrangian
Track sparse features and amplify trajectories
Linear Eulerian
×a
Operate on pixel intensities using spatiotemporal filtering
Phase-based Eulerian
Phase (φ)
×a
Amplify temporal phase variations in the frequency domain
Deep Learning Based Methods
Single-Domain Focused Magnification
Encoder
Manipulator
Decoder
Learn motion representiations in a single domain and amplify in an end-to-end manner
Multi-Domain Synergistic Magnification
Disentangle and synergize features across multiple domains for robust magnification
Applications of Motion Magnification
Visual Vibration Measurement
Amplify subtle vibrations in videos for non-contact, high-precision vibration analysis.
Structural Health Monitoring
Detect and localize structural damage by magnifying small deformation signals.
Biomedical Science
Reveal imperceptible physiological motions for non-invasive diagnosis and monitoring.
Deepfake Detection
FAKE
Magnify subtle inconsistencies in facial dyamics to identify deepfake videos.

tion" OR "Lagrangian motion magnification" OR "Eulerian motion magnification" OR "phase-based motion magnification"). This query was further complemented with auxiliary terms such as ("deep learning", "vibration", "biomedical", and "micro-expression") to extend coverage toward domain-specific applications.

The overall screening pipeline is illustrated in Figure 2. In the identification stage, a total of 404 records were retrieved from the selected databases. During the initial screening phase, 142 records were excluded based on predefined criteria, including duplicate entries, non-English publications, patents, non-peer-reviewed preprints, and studies not directly related to motion magnification (e.g., general optical flow or large-scale motion tracking).

Subsequently, 114 full-text articles were assessed for eligibility, with evaluation criteria focusing on methodological completeness, relevance to motion magnification, and the availability of quantitative validation. Studies lacking sufficient technical detail, experimental support, or clear methodological contribution were excluded at this stage. Ultimately, 64 representative works were retained for comprehensive analysis and comparative synthesis in the following sections.

cations lacking sufficient technical detail; and (3) duplicate or preliminary versions of already published work.

Representative studies were selected based on a combination of methodological novelty, citation impact, and relevance to key application domains. To further mitigate potential selection bias, results from multiple databases were cross-checked, and preference was given to widely adopted, well-documented, or experimentally validated approaches.

Based on the screened literature, Figure 3 illustrates the development timeline of motion magnification. Since its initial introduction around 2005, the field has undergone steady evolution. Prior to 2018, research was predominantly dominated by handcrafted signal-processing approaches. With the emergence of deep learning-based frameworks, the field has experienced rapid expansion in recent years.

Through this structured investigation, existing motion magnification approaches are categorized into two primary groups: conventional methods and deep learning-based methods. A comparative overview of these methodologies is presented in Figure 4. As illustrated, traditional methods rely on handcrafted representations and predefined filters, whereas deep learning-based approaches learn motion representations directly from data, enabling more adaptive and scalable magnification

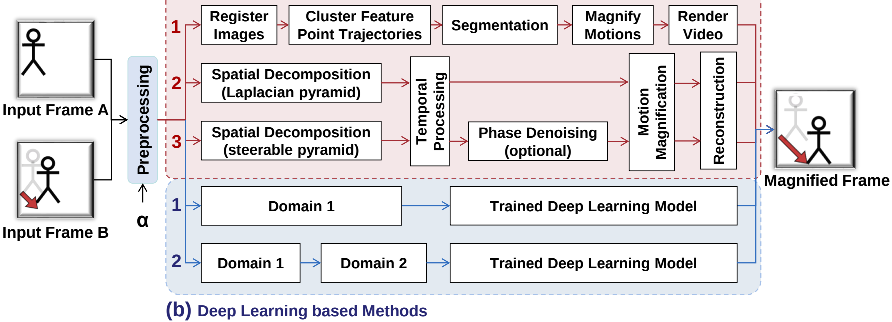


Fig. 4 Motion magnification flowcharts. (a) Traditional methods: (1) Lagrangian, (2) Linear Eulerian, (3) Phase-based Eulerian. (b) Deep learning-based methods: (1) Single-Domain focused, (2) Multi-Domain synergistic.

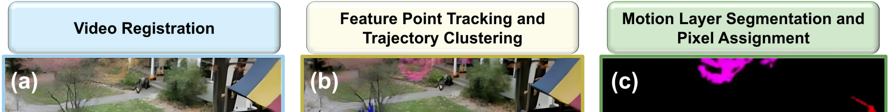


image registration or alignment correction is applied to the input video frames to remove unwanted motion caused by camera shake or slight movements. This process is typically accomplished using techniques like feature point matching[54] and optical flow methods[55–57], which establish pixel correspondences between adjacent frames. This step is essential to stabilize the video, ensuring that subsequent processing focuses solely on the targeted motions, thereby enhancing accuracy and reliability in motion analysis.

II. Feature Point Tracking and Trajectory Clustering

Drawing on the Lagrangian perspective, this approach involves selecting and tracking a series of prominent feature points (such as corners and edges) within the video sequence. These feature points are chosen for their high stability and distinctiveness, ensuring they can be reliably identified across frames. By calculating the displacement of these points in consecutive frames, their motion trajectories are obtained, providing a detailed representation of the movement within the video. This tracking process enables a precise analysis of the dynamic behavior of the observed features.

After extracting suitable feature points, they are then grouped based on the similarity of their motion trajectories, such as speed and direction. Clustering algorithms like K-means[58] and DBSCAN[59] are commonly used for this task, as they effecand noticeable "holes," which are visually disruptive and fall short of human visual standards. To enhance the visual quality, a rendering process is applied. The "holes" can be corrected through techniques like texture synthesis and image interpolation, filling in missing information for a seamless appearance. For the distortion, operations such as noise reduction, contrast enhancement, and color correction are necessary to restore a more natural look, ensuring that the final image is both visually coherent and detailed.

From the above discussion, it is clear that the Lagrangian method is computationally intensive and time-consuming. Its processing steps are tightly interconnected, often requiring multiple manual attempts to select optimal parameters, such as for feature point clustering and motion layer segmentation. Additionally, different scenes necessitate manual parameter adjustments, making it a labor-intensive process to achieve satisfactory magnification results—especially when dealing with long videos or high-resolution data[63–65]. These limitations have hindered the method's further development and widespread application. However, this groundbreaking research has laid a solid foundation for subsequent motion magnification techniques and has demonstrated notable potential for application in fields like structural health monitoring and medical imaging.

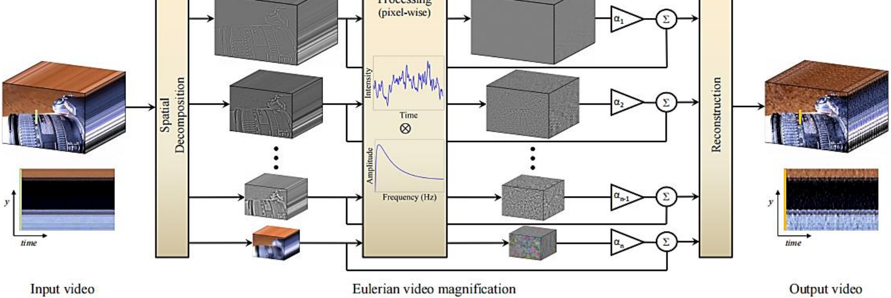


Fig. 6 Overview of the Eulerian video magnification framework[23]. The system employs a three-stage video processing framework: (1) spatial frequency decomposition of the input video, (2) uniform temporal filtering across all frequency bands followed by α-factor amplification, and (3) signal reconstruction through band recombination. This architecture allows flexible parameter optimization, where both temporal filtering characteristics and amplification coefficients can be adaptively configured for specific application scenarios.

To provide an intuitive and mathematically tractable demonstration of the relationship between temporal filtering and motion scaling, we simplify the generalized by the magnification factor α and added back to the original observed signal $I(x,t)$, yielding the processed intermediate signal:

$$\tilde{I}(x,t) = I(x,t) + \alpha B(x,t) \tag{4}$$

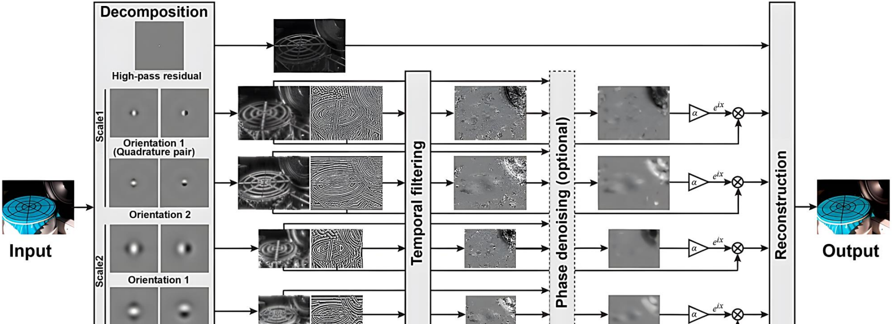
By substituting Equations (2) and (3) into Equation (4), we derive the linearized representation of the synthesized signal:

$$\tilde{I}(x,t) \approx f(x) + (1+\alpha)\delta(t)\frac{\partial f(x)}{\partial x} \tag{5}$$

Under the assumption that the first-order Taylor series expansion remains valid for the augmented displacement, Equation (5) can be mapped back into the spatial domain via an inverse Taylor approximation:

$$\tilde{I}(x,t) \approx f(x + (1+\alpha)\delta(t)) \tag{6}$$

Equation (6) mathematically demonstrates that the subtle temporal displacement each coefficient is characterized by a local amplitude and a local phase. A temporal band-pass filter is subsequently applied to the time-varying phase of each sub-band to isolate the desired motion frequencies. The isolated phase perturbation is then amplified by a specified factor and linearly recombined with the original phase profile. Consequently, in the final synthesized video, the underlying motion-encoded phase is magnified while the local amplitude remains unaltered, as illustrated in Figure 7.



within a specific temporal frequency band, a DC-balanced temporal band-pass filter is applied to the time-varying phase $\omega(x + \delta(t))$. Assuming the motion spectrum lies entirely within the filter's passband, the static spatial component ωx is effectively suppressed, yielding the filtered phase signal:

$$B_\omega(x, t) = \omega\delta(t) \tag{9}$$

To construct the motion-magnified sub-band, the band-passed phase $B_\omega(x, t)$ is scaled by the magnification factor α and added to the phase of the original sub-band $S_\omega(x, t)$:

$$\hat{S}_\omega(x, t) = S_\omega(x, t)e^{i\alpha B_\omega(x,t)} = A_\omega e^{i\omega(x+(1+\alpha)\delta(t))} \tag{10}$$

The effective displacement within the modified sub-band $\hat{S}_\omega(x, t)$ is now linearly scaled to $(1 + \alpha)\delta(t)$. Finally, summing the modified components across all spatial frequencies reconstructs the motion-magnified sequence $\hat{I}(x, t) = f(x + (1 + \alpha)\delta(t))$, thereby completing the inverse pyramid synthesis.

In video signal processing, noise presents a formidable challenge, particularly due to its disruptive impact on intensity and amplitude profiles. Random noise fluctuations often mask subtle, low-amplitude motion signals, severely undermining measurement tude) constitutes the core signal-processing mechanism behind PEVM's exceptional robustness.

Despite the superior noise resilience and visual quality offered by PEVM, its practical execution involves substantial computational trade-offs when contrasted with LEVM and Lagrangian alternatives. Mathematically, LEVM achieves a strict linear time and space complexity of $\mathcal{O}(N)$, where N denotes the total number of pixels within a frame. This linear scaling yields extreme computational efficiency, rendering real-time execution on standard CPU architectures highly feasible. In contrast, the complex steerable pyramid decomposition utilized in PEVM demands $\mathcal{O}(K \cdot L \cdot N)$ space to store complex coefficients across L scales and K orientations, while its temporal complexity—primarily driven by multi-scale Fast Fourier Transforms (FFTs)—reaches $\mathcal{O}(K \cdot L \cdot N \log N)$. This heightened memory footprint and computational overhead create a notable bottleneck for high-resolution video streams, typically necessitating parallelized GPU acceleration to unlock real-time processing.

Limitations notwithstanding, these Eulerian operational profiles remain significantly more efficient than Lagrangian methods, which incur a polynomial complexity ranging from $\mathcal{O}(N \log N)$ to $\mathcal{O}(N^2)$ to compute dense optical flow fields. Consequently, while Lagrangian tracking offers fine grained trajectory precision, its poor scalability

Table 1 Taxonomy, Characteristics, and Functional Roles of Representative MM Datasets.

Dataset	Data Type	Ground Truth (GT)	Primary Role	Typical Scenario / Key Characteristics
Oh et al. [33]	Synthetic	Pixel-level motion field	Training & Testing	Multi-object non-rigid 2D warping; 200k backgrounds / 7k foregrounds.
AxialMM Dataset [78]	Synthetic	Directional displacement field	Training & Testing	Directionally decoupled harmonic motion along specified X/Y axes.
DIS5K-StickPNG [40, 41]	Synthetic	Multi-level synthetic annotations	Robustness Test	High-resolution background textures with injected Poisson noise and Gaussian blur.
Real-world Benchmark Dataset [23, 32, 33]	Real-world	None	Validation & Demo	Consolidated standard suite: *Static Mode* (Baby, Guitar, Crane) and *Dynamic Mode* (Face, Swing, Gunshot).

- **Foundational Synthetic Configurations (Primary Supervised Training):**

Fig. 8 Example frames from dataset. (a) Pasted foreground objects using segmentation from PASCAL VOC and background from MS COCO dataset. (b) Only has background moving to ensure the network learns global motion (c) Background blurred to ensure low contrast texture is well-represented. And another two parts with the same specification as (b) and (c) with background motion removed so that the network learns the changes that are due to noise only.The dataset is available at this link.

Table 2 Systematic Categorization of Evaluation Indicators and Targeted Physical Dimensions.

Targeted Dimension	**Primary Indicators**	**Reference Mode**	**Physical Assessment Goal**
Spatial Fidelity & Clarity	MSE [82], PSNR [82], SSIM [83], LPIPS [84]	Full-Reference	Pixel reconstruction accuracy and structural/perceptual alignment.
Motion Amplitude Accuracy	Temporal slice track (X-T/Y-T), Error Margin	Full / Implicit Ref	Quantifies if output displacement matches the targeted scale ($\alpha \cdot \delta$).
Temporal Consistency	Temporal wrap loss, Inter-frame variance	Full-Reference	Detects flickering artifacts, jittering, or non-uniform frame propagation.
Frequency Fidelity	Power Spectral Density (PSD), Fourier Error	Implicit Reference	Verifies if the amplified frequency matches the source vibration spectrum.
Environmental Perturbations	MUSIQ [85], Hyper-IQA [86], MANIQA	No-Reference	Evaluates sensitivity to photon noise, shadows, and illumination changes.
Computational Efficiency	Inference Latency, FPS, FLOPs, Param Count	Reference-free	Benchmarks deployment viability for edge computing and real-time processing.

features extracted by CNNs trained on the Berkeley-Adobe Perceptual Patch Similarity (BAPPS) dataset, capturing subtle differences and reflecting human perceptual characteristics under high magnification factors.

(b) Motion Amplitude Accuracy: This indicator quantifies whether the actual physical displacement of an object in the magnified output closely matches the targeted mathematical scaling factor ($\alpha \cdot \delta$). On synthetic configurations, this is frequently computed by evaluating the error margin against the explicit ground truth warping field. On real-world sequences lacking explicit fields, researchers typically utilize subpixel tracking along temporal spatial-temporal slices (X-T or Y-T plots) or cross-reference the output displacement with the physical measurements of available laser displacement sensors [78].

(c) Temporal Consistency: Temporal consistency ensures that the magnified video sequence is smooth and free from inter-frame flickering artifacts or non-uniform artificial acceleration. Handcrafted Eulerian networks achieve this via strict temporal bandpass filtering, whereas deep neural networks optimize this dimension by integrating dynamic filters or inter-frame shape difference regularizations to penalize erratic phase variations across consecutive frames.

(d) Frequency Fidelity: Crucial for engineering diagnostics and health moniboundary artifacts, while specialized lightweight architectures explicitly incorporate parameter-efficient scaling or differentiable architecture searches to minimize inference delay for practical deployment.

2.3.2 Single-Domain Focused Magnification

Traditional motion magnification methods rely on manually designed features and filters, which often exhibit limited robustness in complex or non-rigid motion scenarios and are sensitive to noise and illumination variations. With the rapid advancement of computational hardware and deep learning techniques, data-driven approaches have emerged, enabling more accurate and stable motion amplification through end-to-end learning frameworks.

To provide a structured and interpretable overview of deep learning-based methods, we categorize existing approaches into two representative paradigms: Single-Domain Focused Magnification (SDFM) and Multi-Domain Synergistic Magnification (MDSM). Here, the term "domain" refers to either (1) a mathematical representation space (e.g., spatial, frequency, or phase) or (2) a sensing modality (e.g., RGB frames or event streams).

amplification, it is classified as **MDSM** (e.g., MD-VMM [37]).

- **Rule 3 (Multi-Modal Integration):** If the architecture incorporates heterogeneous sensing modalities (e.g., combining RGB frames with event-based signals) and performs joint motion reasoning across modalities, it is classified as **MDSM** (e.g., EB-VMM [42]).

To provide a representative blueprint of the current literature, Table 3 presents a structured technical taxonomy and technical comparison mapping prominent deep learning methods across both paradigms based on their core architectural characteristics. For conceptual and architectural clarity, Fig. 9 highlights the fundamental distinction between SDFM and MDSM in terms of representation space and amplification mechanism, whereas Fig. 10 provides a structural comparison of representative network architectures across the two paradigms.

Table 3 Refined taxonomy and hardware-software configuration matrix of prominent le magnification methodologies.

Method	Input Type	Core Architecture	Loss Functions	Primary Datasets	Re

structed by applying random local pixel displacements, affine transformations, or 2D sinusoidal vibrations to high-quality static images to establish clean ground-truth reference frames with specified magnification factors (α). LN-VMM [38] improves on this by introducing high-throughput training via lightweight surrogate models. For MDSM, MD-VMM [37] and FD-VMM [41] employ frequency-specialized synthetic sets containing decoupled low-frequency spatial structures and high-frequency motion details. Conversely, EB-VMM [42] represents a unique self-supervised paradigm trained on the multi-modal Ev-Magnify dataset, utilizing event-reconstruction consistency (L_{rc}) instead of explicit frame-level ground-truths.

In terms of quantitative assessment and computational complexity, PSNR and SSIM serve as the universal standard evaluation metrics across all seven frameworks to benchmark structural fidelity and noise resistance. Additional metrics include the axial motion score for LBA-VMM [35], temporal consistency indices for STB-VMM [36], and phase-SNR for MD-VMM [37]. Computationally, the models diverge significantly: LN-VMM [38] and LB-VMM [33] deliver real-time processing capabilities with low computational overhead due to their streamlined CNN structures. In contrast, the Swin-Transformer backbone of STB-VMM [36] and the forward/inverse spectral transforms of MD-VMM [37] demand heavy GPU memory allocation and suffer from

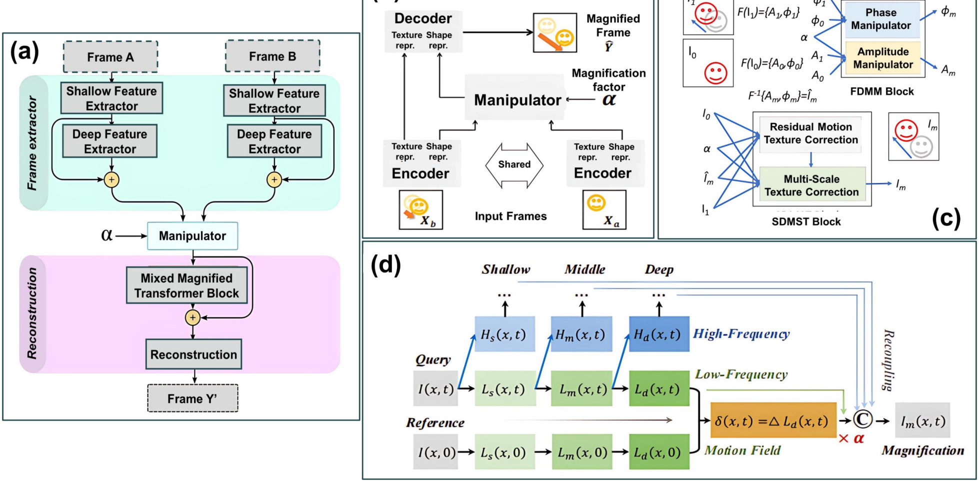


Fig. 10 The network framework of some classical deep learning methods: (a) shows the STB-VMM's[36] network; (b) shows the LB-VMM's[33] network; (c) shows the MD-VMM's[37] network; (d) shows the FD-VMM's[41] network.

scenarios, the authors incorporate localized non-linear mapping operations to suppress distortion, formulated as:

$$G_m(M_a, M_b, \alpha) = M_a + h(\alpha \cdot g(M_b - M_a)), \tag{17}$$

where $g(\cdot)$ and $h(\cdot)$ denote standard convolutional steps and non-linear activation functions, respectively.

During the network training phase, optimization is guided strictly by an L_1 reconstruction loss penalizing the divergence between the predicted output $\hat{Y}$ and the ground-truth magnified frame Y. Empirical evaluations by the authors demonstrated that integrating more sophisticated objectives, such as perceptual loss [89] or adversarial loss [90], yielded negligible performance improvements. From an optimization perspective, the constant gradient magnitude of the L_1 norm ensures robust, linear convergence and mitigates gradient explosion risks. Conversely, the highly non-convex loss surfaces characteristic of adversarial and perceptual objectives tend to destabilize sub-pixel feature manipulation, frequently introducing high-frequency geometric artifacts rather than enforcing strict structural fidelity. To enforce the explicit latent separation between texture and shape representations, a specialized perturbation their architecture effectively isolates and magnifies subtle structural displacements along user-specified coordinate axes, drastically improving motion legibility in complex mechanical systems. Mechanistically, LBA-VMM introduces directional spatial filtering priors to decouple orthogonal noise. While this directional assumption preserves highly sharp boundaries along specified axes, the network sacrifices real-time throughput (~15 FPS) due to the multi-stage directional projections. Its ultimate failure mode occurs when processing chaotic, non-rigid structural fields, where the rigid linear directional assumption breaks down completely and yields severe geometric ghosting artifacts.

Addressing the high computational footprint required for deployment on edge devices, Singh et al. [38] developed a lightweight, real-time alternative (LN-VMM) utilizing an aggressive feature-sharing encoder configuration. By pairing an appearance preservation stream with an explicit edge-preservation loss (L_e), their framework scales down parameter overhead significantly to achieve outstanding real-time edge processing (~90 FPS). The foundational assumption of LN-VMM is that spatial structure boundaries can be explicitly regulated via surrogate boundary constraints. Although the network maintains exceptionally sharp object edges due to the specialized edge loss, its compact parameter capacity bounds its latent representation tional efficiency and noise handling. This lightweight design operates 2.7 times faster with 4.2 times fewer FLOPs than the baseline, enabling real-time 1080p inference without sacrificing generation quality.

Despite these rapid architectural variations, the single-Domain paradigm retains inherent structural constraints. The clear advantage of SDFM frameworks lies in their implementation simplicity and low computational latency, rendering them highly effective for isolated, well-defined 2D spatial displacements. However, because these methods operate monolithically within homogeneous coordinate spaces, they lack the capacity for explicit frequency or multimodal feature decoupling. Consequently, when deployed in complex real-world scenes characterized by low signal-to-noise ratios (SNR), wideband environmental interference, or non-rigid structural deformations, SDFM models frequently suffer from severe texture blurring, geometric distortion, and noise amplification, marking a boundary that necessitates the transition to Multi-Domain architectures.

2.3.3 Multi-Domain Synergistic Magnification

To alleviate the limitations of single-domain representations, recent deep learning-such as temporal inconsistency or spectral leakage.

Beyond mathematical domains, MDSM also extends to multi-modal sensing (Rule 3). EB-VMM [42], for instance, combines RGB frames with event-based data to exploit the high temporal resolution of event cameras. This design enables more accurate capture of subtle temporal variations and alleviates limitations of frame-based sampling.

However, event-based methods depend on sensor-specific characteristics, such as contrast thresholds, which may lead to missing information in low-motion or static regions. In addition, cross-modal alignment introduces additional computational complexity.

Other approaches, such as σ-θ Net [94], attempt to achieve domain disentanglement within a learned representation space without explicit reliance on external modalities. By separating scale and directional components, these methods provide a lightweight alternative, although their performance may degrade in highly irregular motion scenarios.

Overall, MDSM methods demonstrate improved robustness in motion disentanglement and noise suppression compared to single-domain approaches. However, these advantages often come at the cost of increased model complexity, higher computational ing for analytical formulations. SDFM approaches typically operate within a single representation space with implicit disentanglement, while MDSM methods introduce explicit transformations or multi-domain fusion to separate motion and appearance components.

Despite these advantages, deep learning models introduce new trade-offs. The lack of explicit physical constraints may reduce interpretability and lead to data-dependent behavior. Common failure cases include temporal flickering in frame-wise models, artifacts introduced by frequency-domain processing, and sensitivity to modality alignment in multi-modal systems. Furthermore, their computational demands can be substantial, posing challenges for real-time or resource-constrained deployment without model optimization.

Consequently, the selection of an appropriate motion magnification paradigm is task-dependent and constrained by the operational setting:

- **Eulerian Methods** are well-suited for resource-constrained environments and periodic motion analysis, where computational efficiency and stable temporal filtering are required.
- **SDFM** approaches provide a balance between efficiency and reconstruction quality,

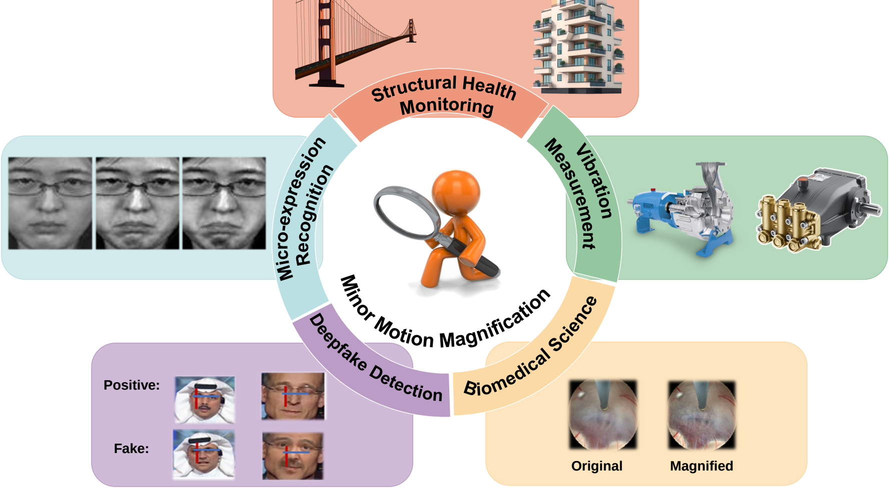


Fig. 11 Empirical taxonomy of motion magnification workflows across diverse analytical fields, featuring representative implementations in visual vibration measurement [44, 94, 95], structural health monitoring [48, 96, 97], biomedical science [98–100], deepfake detection [101–103], and micro-

Table 4 Empirical mapping of typical technical profiles across prominent motion magnification application domains.

Domain	Typical Input Data	Target Scale	Validation Method	Primary Indicators	Common Constraints
Visual Vibration Measurement	High-speed RGB/-Mono ($>$ 250 fps); Macro lens	Sub-pixel to sub-mm (10^{-3} to 10^{-1} px)	Laser Doppler Vibrometer comparison	PSD peaks, FRF, Fourier Error	High-frequency jitter; specular reflections
Structural Health Monitoring	Standard 4K/HD RGB (30/60 fps); UAV/Telephoto	Sub-mm to mm structural drift	Accelerometers or strain gauges	ODS, MAC	Wind-induced camera motion; atmospheric shimmer
Biomedical Analysis	RGB (30 fps); Near-Infrared (NIR)	Micro-vessel pulsing; sub-px motion	PPG or ECG reference	BPM error, RR, SNR	Motion artifacts; illumination variability
Micro-Expression Recognition	Facial video (60–200 fps)	Sub-pixel facial muscle motion	FACS annotations	Accuracy, F1-score	Head motion; occlusion
Deepfake Detection	Compressed video (30 fps)	Subtle temporal inconsistencies	Cross-dataset testing	AUC-ROC, error rate	Compression artifacts

on handcrafted or learned features (e.g., LBP-TOP) [106, 127, 128].

A key challenge is the coexistence of micro-expressions with larger head motion or speech-related movement. Without effective motion disentanglement, magnification may introduce artifacts such as edge distortion or texture inconsistency [129]. Parameter tuning, including the magnification factor α, remains application-dependent [130, 131].

3.5 Deepfake Detection

Deepfake detection leverages MM to reveal subtle temporal inconsistencies in synthesized media, including abnormal motion patterns or missing physiological cues [132–135]. Input data typically consist of compressed video streams (e.g., H.264 or AV1), often with varying quality and bitrate.

Evaluation is conducted through cross-dataset testing (e.g., FaceForensics++) using metrics such as AUC-ROC, Equal Error Rate (EER), and classification accuracy [136, 137].

A major limitation arises from lossy compression, which suppresses high-frequency motion signals and may introduce artificial patterns. In such cases, magnification ducing mass?stiffness constraints can restrict magnification to motion patterns that satisfy realistic oscillation behavior, thereby suppressing stochastic noise.

In addition, improving model interpretability is essential for safety-critical applications, such as biomedical monitoring or precision industrial diagnostics. Future work should explore self-explainable mechanisms, including frequency-aware attention maps or phase-consistency fields, to enable practitioners to verify whether amplified signals correspond to genuine physical phenomena rather than algorithmic artifacts.

4.2 Domain Adaptation and Generative Minor Motion Synthesis

Another major limitation arises from the gap between synthetic training data and real-world deployment environments. Due to the difficulty of acquiring ground-truth minor motion in natural settings, most existing methods rely on synthetic datasets generated through simplified transformations, such as linear 2D warping of static images. While effective for learning basic motion representations, these datasets fail to capture the complexity of real-world conditions, including non-rigid deformation, texture variability, illumination changes, and compression artifacts.

improved efficiency.

In addition, mixed-precision quantization tailored for sub-pixel motion representation can reduce computational cost while preserving numerical stability. Achieving an effective balance between efficiency and accuracy is critical for enabling real-time MM systems on edge devices, such as portable inspection platforms or embedded structural monitoring systems.

5 Conclusion

This review presents a structured and comprehensive synthesis of *minor motion magnification* (MM, also referred to as video motion magnification, MM), with the primary objective of clarifying its theoretical foundations, methodological evolution, and practical deployment characteristics. Specifically, this work establishes a unified mathematical formulation that abstracts diverse approaches under a common reconstruction objective, and proposes a two-tier taxonomy that categorizes deep learning-based methods into Single-Domain Focused Magnification (SDFM) and Multi-Domain Synergistic Magnification (MDSM). In addition, a systematic organization of application domains, datasets, and evaluation protocols is provided to support

Although this work is a review article and does not introduce new experimental models or datasets, we emphasize reproducibility and transparency in the literature screening and taxonomy construction process.

The literature search strategy, including database sources, search date, and query formulation, as well as the inclusion and exclusion criteria, are explicitly described in Section 1. These details enable other researchers to replicate the retrieval procedure under similar conditions.

The representative studies analyzed in this review are reflected in the reference list and are discussed throughout the manuscript, ensuring traceability of the reviewed evidence.

Due to copyright restrictions and database access limitations, the complete set of initially retrieved records and intermediate screening logs cannot be publicly redistributed. However, the methodological transparency provided in this study supports reproducibility at the procedural level.

No new code, pretrained models, or curated datasets were generated in this study. Future work will aim to further enhance openness by releasing structured benchmarks, standardized evaluation protocols, and curated resources where feasible.

sion, Project administration, Resources. **Bo Peng:** Investigation, Formal analysis. **Ningsheng Liao:** Supervision, Writing-Review & Editing.

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